SIGNALS & SPECTRA

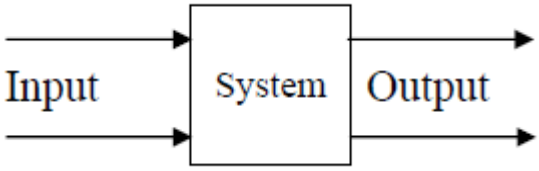
The use of Digital Signal Processing (DSP) is growing exponentially and we have seen its application in control and power systems, biomedical engineering, instrumentation, automotive engineering, telecommunications, mobile communication, speech analysis and synthesis, audio and video processing, seismic, radar and sonar processing and neural computing. DSP's wide spread application is due to its advantages:

- High reliability and reproducibility
- Flexibility and programmability
- The absence of component drift problems
- Compressed storage facility

SIGNALS

Signals are various voltages and currents in a network and can mathematically be represented as functions of time.

The objective of system analysis is to determine the behavior of the system subjected to a specific input or excitation. It is thus convenient to represent a system schematically using blocks.



There are two types of signals in general:

- **CONTINUOUS/ANALOGUE SIGNALS**
  - The independent variable t (usually standing for time) is continuous and thus $f(t)$ is defined for all t.
  - Examples include a person's body temperature, the voltages and currents in a circuit
- **DISCRETE SIGNALS**
  - Discrete signals $f[n]$, $n \in \mathbb{J}$ are defined only at discrete points and consequently the independent variables take on only a discrete set of values. Note that discrete signals are distinct from digital signals which are obtained by quantizing values taken by the discrete signal in order to represent it with a fixed number of bits.
  - Examples include the prices of stocks printed in the daily newspaper, an hourly recording of a person's body temperature

PERIODICITY

A continuous signal $x(t)$ which has the property where:

$$f(t) = f(t+P), \qquad , P \in \mathbb{J}$$

Is said to be periodic for all t. In essence, a periodic signal has the property that it is unchanged by a time shift P, which is referred to as the signal's period.

THE FAMILY OF FOURIER TRANSFORMS

Any continuous periodic signal could be represented as the sum of properly chosen sinusoidal waves. This was first suggested by Fourier. Although a summation of sinusoids cannot form a signal with a corner, its approximation is good enough that the difference between the two has zero energy. This phenomenon goes by the name of Gibbs Effect.

There is a whole family of the Fourier transform[1] which seeks to represent different types of functions as a sum of sinusoidal waves. They are categorized here in terms of their applications

APERIODIC-CONTINUOUS

These signals extend to both positive and negative infinity *without* repeating in a periodic pattern. The Fourier Transform for this type of signal is simply called the **Fourier Transform**.

PERIODIC-CONTINUOUS

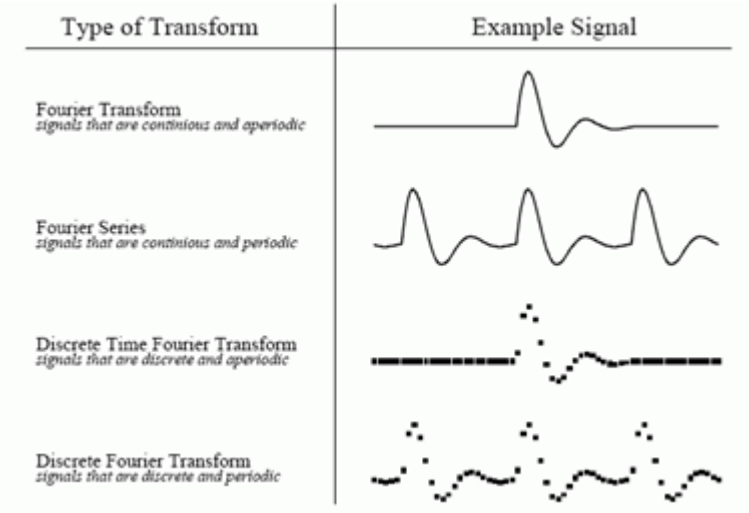
Here the examples include: sine waves, square waves, and any waveform that repeats itself in a regular pattern from negative to positive infinity. This version of the Fourier transform is called the **Fourier Series**.

APERIODIC-DISCRETE

These signals are only defined at discrete points between positive and negative infinity, and do not repeat themselves in a periodic fashion. This type of Fourier transform is called the **Discrete Time Fourier Transform (DTFT).**

PERIODIC-DISCRETE

These are discrete signals that repeat themselves in a periodic fashion from negative to positive infinity. This class of Fourier Transform is most often called the **Discrete Fourier Transform (DFT).**



To rectify the issue of having only a finite length signal (with finite samples) we must make the data look like an infinite length signal by imagining that the signal extends to the left and right of the actual points. If they re valued at zero, then the signal looks discrete and aperiodic – this calls for the DTFT to be applied. If the signal is duplicated to the left and right, then the signal looks discrete and periodic, and the DFT is used.

As it turns out, an *infinite* number of sinusoids are required to synthesize a signal that is *aperiodic*. This makes it impossible to calculate the Discrete Time Fourier Transform in a computer algorithm. By elimination, the only type of Fourier transform that can be used in DSP is the DFT. We thus see that computers can only work with signals that are discrete and finite in length.

[1] A transformation is a function mapping a set X onto another set or onto itself.

We shall see that:

- Both the Fourier transform and DTFT produce continuous spectra.
- The DFT transforms a discrete signal onto a discrete spectrum.
- The DFT coefficients are complex and are typically represented as magnitude and phase spectra.
- The DFT coefficients do not have frequencies that are integer multiples of a fundamental frequency.

FOURIER SERIES: CONTINUOUS TIME PERIODIC SIGNALS

According to the Fourier Theorem, any practical periodic function of fundamental frequency $\omega_0 = 2\pi f_0$ rad/s can be expressed as an infinite sum of sinusoidal functions that are integral multiples of ω_0. We thus arrive at the fourier series:

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

The Fourier series of a periodic function $f(t)$ is a representation that resolves $f(t)$ into a DC component and an AC component comprising an infinite series of harmonic sinusoids.

To find the Fourier coefficients:

$$a_0 = \frac{1}{T}\int_0^T f(t)dt$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\omega_0 t)\ dt$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin(n\omega_0 t)\ dt$$

A Fourier Series may also be represented in amplitude phase form:

$$f(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n)$$

Where:

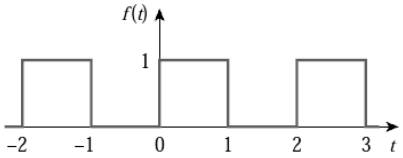
$$A_n = \sqrt{a_n^2 + b_n^2}$$

$$\phi_n = -\tan^{-1}\frac{b_n}{a_n}$$

In general, the Fourier series expansion of a periodic signal will have infinitely many harmonics, except in certain circumstances in which some of the coefficients are zero.

Example

Determine the Fourier series of the waveform shown.



$$a_0 = \frac{1}{T}\int_0^T f(t)dt = \frac{2}{2}\left[\int_0^1 1\,dt + \int_1^2 0\,dt\right]$$
$$= \frac{1}{2}$$

Now: $\omega_0 = 2\pi F_0 = \frac{2\pi}{2} = \pi$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\pi t)\,dt = \int_0^1 \cos(n\pi t)\,dt + \int_1^2 0\,dt$$
$$= \frac{1}{n\omega_0}[\sin n\pi t]_0^1$$
$$= 0$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin(n\pi t)\,dt = \int_0^1 \sin(n\pi t)\,dt + \int_1^2 0\,dt$$

$$= \left[-\frac{1}{n\pi}\cos(n\pi t)\right]_0^1$$

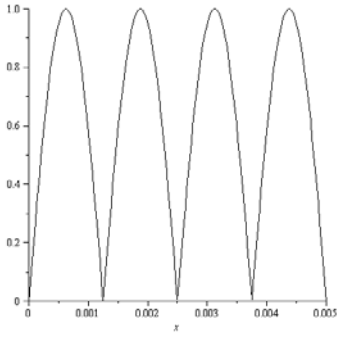
$$= -\frac{1}{n\pi}(\cos(n\pi) - 1)$$

$$= \begin{cases} \frac{2}{n\pi} & , \quad n = odd \\ 0 & , \quad n = 0 \end{cases}$$

Example

Given that $f_1 = 400\ Hz$. Find the Fourier Series coefficients of $x(t) = |\sin(2\pi f_1 t)|$

This wave form is a full rectified sine wave:



We consider the first half of the sine wave to be the full period waveform so: $T = \frac{1}{800}$, $\omega_0 = 1600\pi$. Since this is an even function we have $b_n = 0$. Thus: $c_n = \frac{a_n}{2}$:

$$a_n = \frac{2}{T}\int_0^T x(t) cosn\omega_0 t\ dt$$

$$= 1600\int_0^{\frac{1}{800}} \sin(800\pi t)\cos(1600\pi n t)\ dt$$

But by the auxiliary angle results:

$$\sin(A+B) + \sin(A-B) = 2sinAcosB$$

$$\begin{aligned} a_n &= 800\int_0^{\frac{1}{800}} \sin(800\pi t[1+2n]) + \sin(800\pi t[1-2n])\ dt \\ &= \frac{1}{\pi(1+2n)}(-cos\pi(1+2n)+1\ ) + \frac{1}{\pi(1-2n)}(-cos\pi(1-2n)+1) \\ &= \frac{2}{\pi(1+2n)} + \frac{2}{\pi(1-2n)} \\ &= \frac{2(1-2n)+2(1+2n)}{\pi(1-4n^2)} \\ &= 4/\pi(1-4n^2) \end{aligned}$$

FREQUENCY SPECTRUM

A periodic signal $x(t)$ is repetitive with n multiples of the fundamental period T giving $x(t) = x(t+T)$. This creates a dilemma in which we can no longer refer to the signal $x(t)$ as having a frequency of $\frac{1}{T}$ Hz because the Fourier series suggests that it is composed of many harmonics, each of a different frequencies.

The set of frequencies present in a given periodic signal is the **spectrum of the signal**. The spectrum is normally infinitely large, but it may be the case that the coefficients are zero for some $|n| > N$, in which case the signal is said to be **bandlimited** between $-\frac{N}{T}\ Hz$ and $\frac{N}{T} Hz$.

The (discrete) frequency spectrum of a signal consists of the plots of the amplitudes and phases of the harmonics versus frequency.

The plot of the amplitude A_n of the harmonics versus $n\omega_0$ is called the amplitude spectrum of $f(t)$, whereas a plot of the phase ϕ_n versus $n\omega_0$ is the phase spectrum of $f(t)$

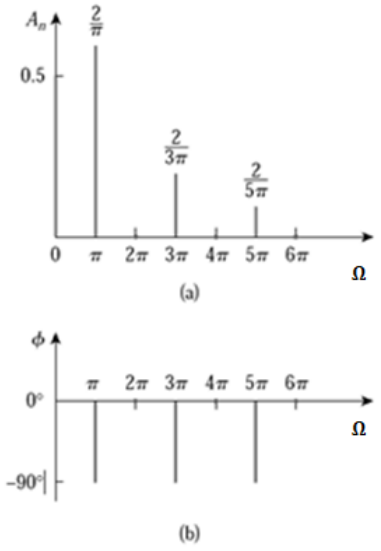
Example

Obtain the amplitude and phase spectra for the (rectangular) pulse train.

The amplitude-phase form of Fourier series is more appropriate:

$$A_n = \sqrt{a_n^2 + b_n^2} = \begin{cases} \frac{2}{n\pi}, & n = odd \\ 0, & n = even \end{cases}$$

$$\phi_n = -\tan\left(\frac{b_n}{a_n}\right) = \begin{cases} -\frac{\pi}{2}, n = odd\ (n \to \infty) \\ 0, n = even \end{cases}$$



Symmetry Considerations

Even Symmetry

When a function $f(t)$ is even it is symmetrical about the vertical axis: $f(t) = f(-t)$. Then we have:

$$a_0 = \frac{2}{T}\int_0^{\frac{T}{2}} f(t)\,dt$$

$$a_n = \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\cos(n\omega_0 t)\;dt$$

$$b_n = 0$$

Odd Symmetry

A function $f(t)$ is said to be odd if its plot is asymmetric about the vertical axis: $f(-t) = -f(t)$

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\sin(n\omega_0 t)\;dt$$

Example

Find the Fourier series of $f(t)$

This is an odd function so $a_0 = a_n = 0$

The period is $T = 4 \rightarrow \omega_0 = \frac{2\pi}{T} = \frac{\pi}{2}$

$$b_n = \int_0^2 f(t)\sin(n\omega_0 t)\;dt = \int_0^1 \sin\left(\frac{n\pi}{2}t\right)\;dt$$

$$= -\frac{2}{n\pi}\left[\cos\left(\frac{n\pi}{2}t\right)\right]_0^1$$

$$= -\frac{2}{n\pi}\left[\cos\left(\frac{n\pi}{2}\right) - 1\right]$$

$$f(t) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\cos\left(1 - \frac{n\pi}{2}\right)\sin\left(\frac{n\pi}{2}t\right)$$

COMPLEX EXPONENTIAL FOURIER SERIES

Using Euler's identity:

$$\cos(n\omega_0 t) = \frac{1}{2}(e^{jn\,\omega_0 t} + e^{-jn\,\omega_0 t})$$

$$\sin(n\omega_0 t) = \frac{1}{2j}(e^{jn\,\omega_0 t} - e^{-jn\,\omega_0 t})$$

By substituting these relationships and gathering like terms we obtain:

$$f(t) = a_0 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n - jb_n)\,e^{jn\,\omega_0 t} + (a_n + jb_n)e^{-jn\,\omega_0 t}$$

We let:

$$c_0 = a_0, \qquad c_n = \frac{a_n - jb_n}{2}, \quad c_{-n} = \frac{a_n + jb_n}{2}$$

Then we get:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\,\omega_0 t}$$

With $c_n = \overline{c_{-n}}$ - complex conjugates. If $c_n = 0$, $|n| > N$ then we see that the signal is bandlimited such that there is a finite number (N) of harmonics. To find the complex Fourier series coefficients we perform the integral:

$$c_n = \frac{1}{T}\int_0^T f(t)e^{-jn\,\omega_0 t}\,dt = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)e^{-jn\,\omega_0 t}\,dt$$

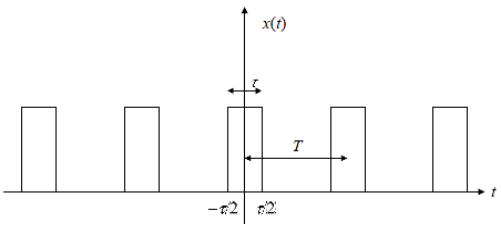
Note that the interval of integration can be from any two points that are one period apart: t_0 to $t_0 + T_p$

The connection between the three forms of presenting Fourier series is given by:

$$A_n\angle\phi_n = a_n - jb_n = 2c_n = 2|c_n|e^{j\,\phi_n}$$

Example

Find the complex Fourier coefficients for the following periodic signal $x(t)$



$$c_n = \frac{1}{T}\int_0^T f(t)e^{-jn\omega_0 t}\,dt = \frac{1}{T}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} 1\cdot e^{-jn\omega_0 t}\,dt$$
$$= -\frac{1}{Tjn\omega_0}\left[e^{-jn\omega_0 t}\right]_{-\frac{\tau}{2}}^{\frac{\tau}{2}}$$
$$= -\frac{1}{Tjn\omega_0}\left[e^{\frac{-jn\omega_0\tau}{2}} - e^{\frac{jn\omega_0\tau}{2}}\right]$$
$$= -\frac{2j}{Tjn\omega_0}\sin\left(\frac{n\omega_0\tau}{2}\right)$$
$$= \frac{\tau}{T}\frac{\sin\left(\frac{n\omega_0\tau}{2}\right)}{\frac{n\omega_0}{2}}$$
$$= \frac{\tau}{T} = sinc\left(\frac{n\omega_0\tau}{2}\right)$$

Note that:

$$sinc(x) = \frac{sinx}{x}$$

$$sinc(0) = 1$$

$$sinc(n\pi) = 0, \qquad n = 1,2,3,\ldots$$

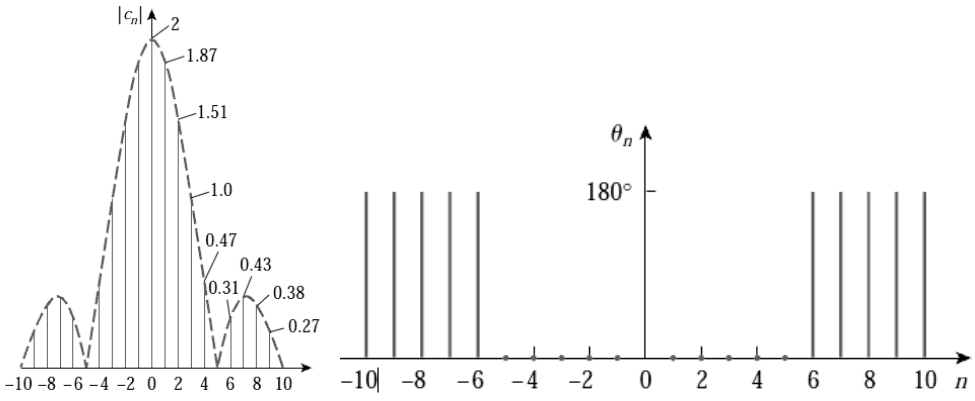
The sinc function also shows even symmetry.

Example

Derive the line spectra for $c_n = 2sinc(\frac{n\pi}{5})$

Since c_n is purely real, then its phase can only be on the horizontal axesz:

$$\phi_n = \begin{cases} 0,\ \sin\left(\frac{n\pi}{5}\right) > 0 \\ 180°, \sin\left(\frac{n\pi}{5}\right) < 0 \end{cases}$$



Example

Given that $f_1 = 400\ Hz$. Find the Fourier Series coefficients of $x(t) = |\sin(2\pi f_1 t)|$ using complex exponentials

$$
\begin{aligned}
c_n &= \frac{400}{j}\int_0^{\frac{1}{800}} \left(e^{j800\pi t} - e^{-k800\pi t}\right) e^{-j1600\pi nt}\ dt \\
&= \frac{400}{j}\int_0^{\frac{1}{800}} e^{j800\pi t(1-2n)} - e^{-j800\pi t(1+2n)}\ dt \\
&= \frac{400}{j}\left[\left(\frac{1}{800\pi j(1-2n)}\right)\left[e^{j800\pi t(1-2n)}\right]_0^{\frac{1}{800}} + \left(\frac{1}{800\pi j(1+2n)}\right)\left[e^{-j800\pi t(1+2n)}\right]_0^{\frac{1}{800}}\right] \\
&= \frac{-1}{2\pi(1-2n)}\left[e^{\pi j(1-2n)} - 1\right] + \frac{-1}{2\pi(1+2n)}\left[e^{-j\pi(1+2n)} - 1\right] \\
&= \frac{-1}{2\pi j(1-2n)}[-2] + \frac{-1}{2\pi j(1+2n)}[-2] \\
&= \frac{+2(1+2n) + 2(1-2n)}{2\pi j(1-4n^2)} \\
&= \frac{2}{\pi(1-4n^2)}
\end{aligned}
$$

1/T could be wrong here because we now treat it as a sine wave with period 1/800.

FOURIER TRANSFORM

The Fourier series enables us to represent a periodic function as the sum of sinusoids and to obtain the frequency spectrum from the series. For signals which are not periodic, we can use the **Fourier transform** to convert a function in the time domain to the frequency domain, even if the function is not periodic.

The Fourier transform pair is given by:

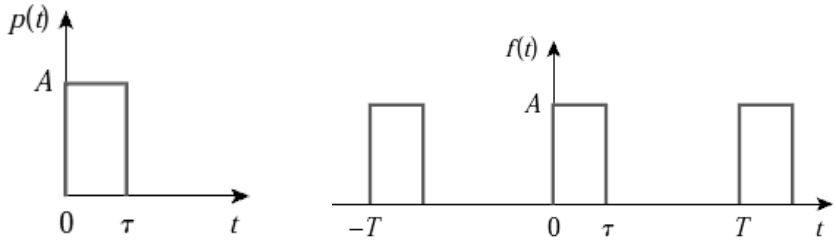
$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\,d\omega \quad \leftrightarrow \quad F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt$$

We can also express the Fourier transform pair in terms of the frequency variable, F, which is in Hertz:

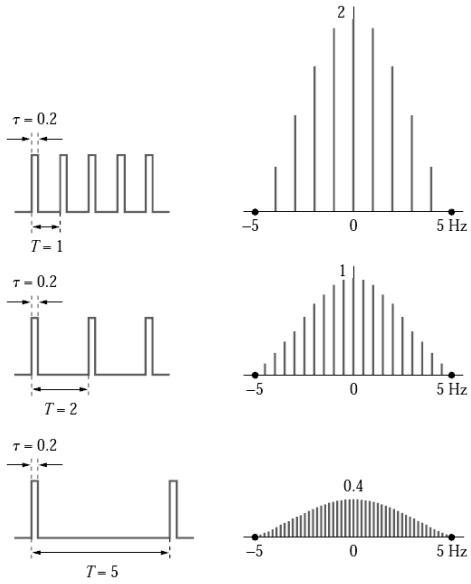
$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(f)e^{j2\pi ft}\,df \leftrightarrow X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\,dt$$

Proof

Consider a nonperiodic function $p(t)$. From this aperiodic signal, we can create a periodic signal $f(t)$ whose shape over one period is the same as $p(t)$ and $T > \tau$.



As $T \to \infty$, we end up with a single pulse of width τ and the function $f(t)$ is no longer periodic. Note that the frequency at which the envelope first becomes zero remains the same.



However, the amplitude of the spectrum and the spacing between adjacent components both decrease, while the number of harmonics increases. This results in the sum of the amplitudes of the harmonics to be almost constant.

As the total energy of the components within a band must remain unchanged, the amplitudes of the harmonics must decrease as T increases and ω decreases until the discrete *spectrum ultimately becomes continuous*.

This means that we can now express $f(t)$ as a Fourier series whose Fourier coefficient is given by: $c_n = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)e^{-j\omega_0 t}\,\mathrm{dt}$.

This gives the complex Fourier series:

$$f(t) = \sum_{n=-\infty}^{\infty} \left[\frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) e^{-jn\,\omega_0 t}\ dt \right] e^{jn\,\omega_0 t}$$

Note that as described above, ω decreases as T increases so:

$$\Delta\omega \to \omega = \frac{2\pi}{T}$$

$$f(t) = \sum_{n=-\infty}^{\infty} \left[\frac{\Delta\omega}{2\pi} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) e^{-jn\,\omega_0 t}\ dt \right] e^{jn\,\omega_0 t}$$

But also note that as $T \to \infty$

$$\sum_{n=-\infty}^{\infty} \to \int_{-\infty}^{\infty}$$

And we find that we will get the continuous variable ω

$$n\omega_0 \to \omega$$

$$\Delta\omega \to d\omega$$

Hence:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) e^{-j\,\omega t}\ dt \right] e^{j\,\omega t}\ dt$$

We define the Fourier transform of $f(t)$ as:

$$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\,\omega t}\ dt$$

As is the case of the Fourier series, $F(\omega)$ is complex valued and can be written as:

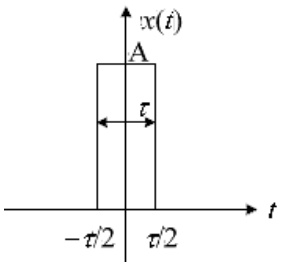
$$F(\omega) = |F(\omega)| e^{j\phi\,(\omega)}$$

As is the case of the Fourier series, the signal is said to be bandlimited, with a bandwidth of $\frac{B}{2\pi}\ Hz$ if:

$$F(\omega) = 0, \qquad |\omega| > B$$

Example

Evaluate the Fourier transform of a rectangular pulse shown below.



$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt = \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} Ae^{-j\omega t}\,dt$$

$$= -\frac{A}{j\omega}\left[e^{-j\omega t}\right]_{-\frac{\tau}{2}}^{\frac{\tau}{2}}$$

$$= \frac{2A}{j\omega}\sin\left(\frac{j\omega\tau}{2}\right)$$

$$= A\tau\frac{\left(\sin\left(\frac{j\omega\tau}{2}\right)\right)}{\frac{j\omega\tau}{2}}$$

$$= A\tau sinc\left(\frac{j\omega\tau}{2}\right)$$

Example

Find the Fourier transform of $\delta(t-t_0)$, $e^{j\omega_0 t}$, $\cos(\omega_0 t)$

a)

$$\mathcal{F}[\delta(t-t_0)] = \int_{-\infty}^{\infty} \delta(t-t_0)e^{-(j\omega t)}\,dt$$

$$= e^{-j\omega t_0}$$

Due to the sifting property of the impulse function.

b)

We let $F(\omega) = \delta(\omega-\omega_0)$ and use the sifting property again:

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(\omega-\omega_0)e^{j\omega t}\,dt = \frac{1}{2\pi}e^{j\omega_0 t}$$

Now that we have found $f(t)$ then we can apply the rule: $F(\omega) = \mathcal{F}[f(t)]$ and note the linearity property which will be discussed in more detail afterwards.

$$\therefore\ \mathcal{F}[e^{j\omega_0 t}] = 2\pi\delta(\omega-\omega_0)$$

c)

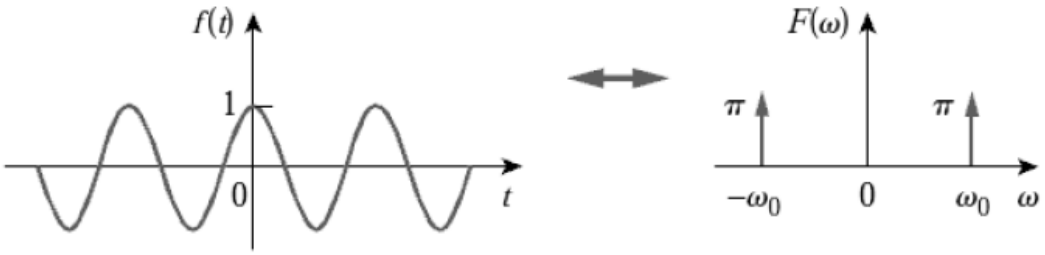
Now: $\cos(\omega_0 t) = \frac{1}{2}[e^{j\omega_0 t} - e^{-j\omega_0 t}]$

$$\mathcal{F}[\cos(\omega_0 t)] = \mathcal{F}\left[\frac{1}{2}[e^{j\omega_0 t} - e^{-j\omega_0 t}]\right]$$

$$= \frac{1}{2}(\mathcal{F}[e^{j\omega_0 t}] - \mathcal{F}[e^{-j\omega_0 t}])$$

$$= \pi\big(\delta(\omega - \omega_0) - \delta(\omega + \omega_0)\big)$$

The resulting spectrum is shown:



CONVOLUTION

Convolution means folding and is an invaluable tool that provides a mean of finding the response $y(t)$ of a system to an excitation $x(t)$, knowing the system impulse response $h(t)$. This is achieved through the convolution integral:

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(t-\lambda)d\lambda = x(t) * h(t)$$

Where λ is a dummy variable and the asterisk denotes convolution. The convolution process is also commutative so $x(t) * h(t) = h(t) * x(t)$.

The **convolution** of two signals consists of time-reversing one of the signals, shifting it, and multiplying it point by point with the second signal, and integrating the product.

The properties of the convolution integral include:

1. Commutative $x(t) * h(t) = h(t) * x(t)$
2. Distributive $f(t) * [x(t) + y(t)] = f(t) * x(t) + f(t) * y(t)$
3. $f(t) * \delta(t) = \int_{-\infty}^{\infty} f(\lambda)\delta(t-\lambda)d\lambda = f(t)$
4. $f(t) * \delta(t - t_0) = f(t - t_0)$
5. $f(t) * \delta'(t) = \int_{-\infty}^{\infty} f(\lambda)\delta'(t-\lambda)d\lambda = f'(t)$
6. $f(t) * u(t) = \int_{-\infty}^{\infty} f(\lambda)u(t-\lambda)d\lambda = \int_{-\infty}^{t} f(\lambda)d\lambda$

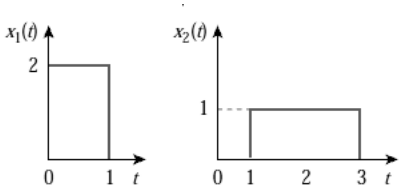
Edmund Li

Convolution performed graphically takes the following steps:

1. **Folding**: Take the mirror image of $h(\lambda)$ or $x(\lambda)$ about the ordinate axis to obtain $h(-\lambda)$ or $x(-\lambda)$
2. **Displacement**: Shift or delay $h(-\lambda)$ or $x(-\lambda)$ by t to obtain $h(t-\lambda)$ or $x(t-\lambda)$
3. **Multiplication**: Find the product of $h(t-\lambda)\cdot x(\lambda)$ or $x(t-\lambda)\cdot h(\lambda)$
4. **Integration**: For a given time t, calculate the area under the product for $0<\lambda<t$ to get $y(t)$ at t.

Example

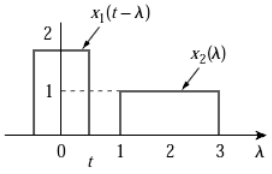
Find the convolution of the two signals.



We first flip $x_1(t)$ and shift it by t to get $x_1(t-\lambda)$ We now multiply the two functions and integrate to determine the area of the overlapping region.

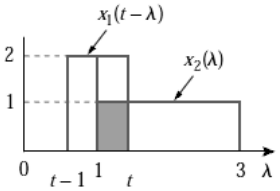
Consider when $0<t<1$. The two functions do not overlap and hence:

$$x_1(t) * x_2(t) = 0$$



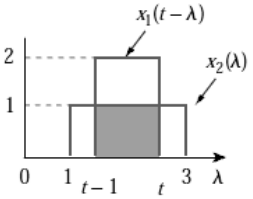
For $1<t<2$, we find that the two functions overlap between 1 and t:

$$y(t) = \int_1^t (2)(1)d\lambda = 2(t-1)$$



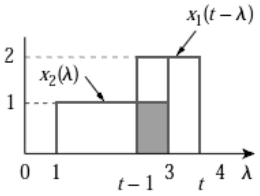
For $2<t<3$, the two signals completely overlap and we get:

$$y(t) = \int_{t-1}^t (2)(1)d\lambda = 2$$



For $3<t<4$, the two signals overlap between $t-1$ and 3.

$$y(t) = \int_{t-1}^3 (2)(1)d\lambda = 2\lambda = 8-2t$$

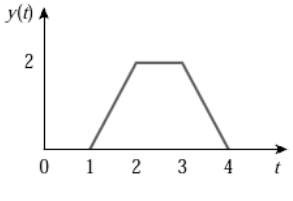


And for $t>4$, the two signals do not overlap so:

$$y(t) = 0$$

We thus obtain:

$$y(t) = \begin{cases} 0, & 0 \le t \le 1 \\ 2t-2, & 1 \le t \le 2 \\ 2, & 2 \le t \le 3 \\ 8-2t, & 3 \le t \le 4 \\ 0, & t \ge 4 \end{cases}$$



PROPERTIES OF THE FOURIER TRANSFORM

LINEARITY

If $F_1(\omega)$ and $F_2(\omega)$ are the Fourier transforms of $f_1(t)$ and $f_2(t)$ respectively:

$$\mathcal{F}[\alpha f_1(t) + \beta f_2(t)] = \alpha F_1(\omega) + \beta F_2(\omega)$$

The same is true for the inverse Fourier transform because it is commutative:

$$\mathcal{F}^{-1}[\alpha F_1(\omega) + \beta F_2(\omega)] = \alpha f_1(t) + \beta f_2(t)$$

Proof:

$$\begin{aligned}\mathcal{F}[\alpha f_1(t) + \beta f_2(t)] &= \int_{-\infty}^{\infty} (\alpha f_1(t) + \beta f_2(t)) e^{-j\omega t}\, dt \\ &= \alpha \int_{-\infty}^{\infty} f_1(t) e^{-j\omega t}\, dt + \beta \int_{-\infty}^{\infty} f_2(t) e^{-j\omega t}\, dt \\ &= \alpha F_1(\omega) + \beta F_2(\omega)\end{aligned}$$

TIME SCALING

If $F(\omega) = \mathcal{F}[f(t)]$ then for *a* being constant:

$$\mathcal{F}[f(at)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

We see that for |a|>1, frequency compression occurs, whereas for |a|<1, frequency expansion occurs.

Proof:

$$\mathcal{F}[f(at)] = \int_{-\infty}^{\infty} f(at) e^{-j\omega t}\, dt$$

Using substitutions, let $x = at, \therefore dx = a\, dt$

$$\begin{aligned}\mathcal{F}[f(at)] &= \frac{1}{a} \int_{-\infty}^{\infty} f(x) e^{-\frac{j\omega x}{a}}\, dx \\ &= \frac{1}{a} F\left(\frac{\omega}{a}\right)\end{aligned}$$

Time Shift

If $F(\omega) = \mathcal{F}[f(t)]$ then

$$\mathcal{F}[f(t - t_0)] = e^{-j\omega t_0}F(\omega)$$

Proof:

$$\mathcal{F}[f(t - t_0)] = \int_{-\infty}^{\infty} f(t - t_0)e^{-j\omega t}\ dt$$

Let $x = t - t_0, \therefore dx = dt$

$$\int_{-\infty}^{\infty} f(x)e^{(-j\omega)(x+t_0)}\ dx = e^{-j\omega t_0}\int_{-\infty}^{\infty} f(x)e^{-j\omega x}\ dx$$
$$= e^{-j\omega t_0}F(\omega)$$

Frequency Shift

If $F(\omega) = \mathcal{F}[f(t)]$ then

$$\mathcal{F}\left[f(t)e^{j\omega_0 t}\right] = F(\omega - \omega_0)$$

Proof:

$$\mathcal{F}[f(t)e^{j\omega_0 t}] = \int_{-\infty}^{\infty} f(t)e^{j\omega_0 t}e^{-j\omega t}\ dt$$
$$= \int_{-\infty}^{\infty} f(t)e^{-j(\omega-\omega_0)t}\ dt$$
$$= F(\omega - \omega_0)$$

Time Differentiation

If $F(\omega) = \mathcal{F}[f(t)]$ then

$$\mathcal{F}[f^n(t)] = (j\omega)^n f(\omega)$$

Proof (for degree 1):

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{-j\omega t}\ d\omega$$

$$f'(t) = j\omega\mathcal{F}^{-1}[F(\omega)] = j\omega f(t)$$

$$\mathcal{F}[f'(t)] = j\omega F(\omega)$$

Differentiation in Frequency

If $F(\omega) = \mathcal{F}[f(t)]$ then

$$\frac{d}{d\omega}F(\omega) = -jtf(t)$$

Convolution in Time

Recall that:

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\lambda)x(t-\lambda)d\lambda$$

Then if $X(\omega), H(\omega), Y(\omega)$ are the Fourier transforms of $x(t), h(t), y(t)$ then:

$$Y(\omega) = \mathcal{F}[h(t) * x(t)] = H(\omega)X(\omega)$$

Which indicates that convolution in the time domain corresponds with multiplication in the frequency domain.

Proof:

$$Y(\omega) = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} h(\lambda)x(t-\lambda)d\lambda\right]e^{-j\omega t}\,dt$$

We now exchange the order of integration since it is a double integral:

$$\int_{-\infty}^{\infty} h(\lambda)\left[\int_{-\infty}^{\infty} x(t-\lambda)e^{-j\omega t}dt\right]d\lambda$$

Now let $\tau = t - \lambda, \;\; d\tau = dt$

$$\int_{-\infty}^{\infty} h(\lambda)\left[\int_{-\infty}^{\infty} x(\tau)e^{-j\omega(\tau+\lambda)}d\tau\right]d\lambda$$

$$= \int_{-\infty}^{\infty} h(\lambda)\left[e^{-j\omega\lambda}\int_{-\infty}^{\infty} x(\tau)e^{-j\omega(\tau)}d\tau\right]d\lambda$$

Again we note that pulling out $e^{-j\omega\lambda}$ will mean that the integral inside the brackets is independent of λ and we just perform the two integrals separately:

$$= \int_{-\infty}^{\infty} h(\lambda)e^{-j\omega\lambda}\,d\lambda\left[\int_{-\infty}^{\infty} x(\tau)e^{-j\omega(\tau)}d\tau\right]$$

$$= H(\omega)X(\omega)$$

Convolution in Frequency

If $X(\omega), H(\omega)$ are the Fourier transforms of $x(t), h(t)$ respectively then:

$$Y(\omega) = \mathcal{F}[x(t)y(t)] = \frac{1}{2\pi}[H(\omega) * X(\omega)]$$

More commonly we express this as:

$$\mathcal{F}[2\pi x(t)y(t)] = H(\omega) * X(\omega)$$

Duality

If $F(\omega)$ is the Fourier transform of $f(t)$ then:

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

Common Fourier Pairs

We present the most commonly used (Continuous Time) Fourier Pairs. As you will see later, there are some strong similarities between these pairs and those of the DTFT.

$f(t)$	$F(\omega)$
$\boldsymbol{\delta(t)}$	1
$\mathbf{1}$	$2\pi\delta(\omega)$
$\boldsymbol{u(t)}$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$\boldsymbol{\lvert t \rvert}$	$-2\omega^2$
$\boldsymbol{e^{j\omega_0 t}}$	$2\pi\delta(\omega - \omega_0)$
$\boldsymbol{e^{-a\lvert t \rvert}}$	$\frac{2a}{a^2 + \omega^2}$
$\boldsymbol{sin\omega_0 t}$	$\pi j[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$\boldsymbol{cos\omega_0 t}$	$\pi(\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$

DISCRETE SIGNALS

We describe a discrete signal that has been derived from taking samples of an analogue signal $x(t)$ at discrete points $t = nT$ as:

$$x[n] \stackrel{\text{def}}{=} x(nT)$$

The constant T referred to as the sampling period whereas the sampling frequency is given by $f_s = \frac{1}{T}\ Hz$ and $\omega_s = 2\pi f_s$ as the sampling angular frequency. We thus have the relationship:

$$t = nT = \frac{n}{f_s} = \frac{2\pi n}{\omega_s}$$

Then for a sinusoid:

$$x(t) = Acos(2\pi f t + \phi)$$

The discrete representation would be:

$$x(nT) = Acos\left(\frac{2\pi f}{f_s} n + \phi\right)$$

$$= Acos(\theta n + \phi)$$

Where $\theta = \frac{2\pi f}{f_s}$ is known as the digital frequency and is defined between:

$$-\pi \leq \theta \leq \pi$$

This equivalently means:

$$-\frac{1}{2} f_s \leq f \leq \frac{1}{2} f_s$$

Since an analog signal can have infinite frequencies, then these frequencies must be mapped between the limits given above. This mapping is known as sampling and requires the sampling frequency to be at least twice that of the maximum frequency present in the analog signal. This is implicitly shown in the above limit.

PERIODIC DISCRETE SIGNALS

So far we have seen that a periodic continuous time signal has the property that $x(t) = x(t+P)$. Similarly, for a discrete periodic signal we have the property:

$$x[n] = x[n+N]$$

Where N is the period length – the number of samples apart in which two similar points are located. If we restrict ourselves to sinusoids (since we can represent any waveform with sinusoids), then we find that:

$$\sin\big((n+N)\theta\big) = \sin(n\theta)$$

Since sinusoids are 2π periodic, we have:

$$N\theta = 2\pi k$$

$$N = \frac{2\pi k}{\theta} = \frac{2\pi k}{\frac{2\pi f_a}{f_s}} = \frac{f_s}{f_a}k$$

When the smallest value of k is selected to ensure that N is an integer, we refer to this particular N as the **fundamental period.**

Example

Determine the fundamental period of $x[n] = 10\cos\left(\frac{2\pi}{15}n + \frac{\pi}{5}\right)$

We saw already that N is $N = \frac{2\pi k}{\theta}$, and thus we can actually express a sinusoid in the form of:

$$x[n] = 10\cos\left(\frac{2\pi}{N}n + \frac{\pi}{5}\right)$$

Thus N=15.

Alternatively, we note that:

$$\theta = \frac{2\pi}{15}$$

$$N = \frac{2\pi}{\frac{2\pi}{15}} = 15$$

Example

Determine the smallest digital frequency given that $N = 10$

$$\theta = \frac{2\pi k}{10}$$

Set k=1 then:

$$\theta = \frac{\pi}{5}\ rad/cycle$$

DISCRETE TIME FOURIER SERIES (DTFS)

Suppose that we are given a periodic sequence $x(n) with\ period\ N$, then we know that $x(n)$ consists of N harmonically related exponential fuctions $e^{\frac{j2\pi kn}{N}}$, giving us the DTFS:

$$x[n] = \sum_{k=0}^{N-1} c_k e^{\frac{j2\pi kn}{N}}$$

With the Fourier coefficients given by:

$$c_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{\frac{-j2\pi kn}{N}}$$

These Fourier coefficients $\{c_k\}$, k=0,1…N-1 provide the description of $x[n]$ in the frequency domain, in the sense that c_k represents the amplitude and phase associated with the frequency component:

$$s_k[n] = e^{\frac{k2\pi kn}{N}}$$

But since the complex exponential is periodic, then so too must the Fourier coefficients c_k satisfy this periodicity condition with which the fundamental period is N:

$$c_{k+N} = c_k$$

Example

Determine the spectra for the signal $x[n] = \{1,1,0,0\}$ with period N=4.

$$c_k = \frac{1}{4}\sum_{n=0}^{3} x[n] e^{\frac{-j2\pi kn}{4}}$$
$$= \frac{1}{4}\left(1 + e^{\frac{-j\pi k}{2}}\right)$$

$$c_0 = \frac{1}{4}, c_1 = \frac{1}{4}(1-j), c_2 = 0, c_3 = \frac{1}{4}(1+j)$$

Thus the magnitude and phase spectra are:

	c_0	c_1	c_2	c_3
$\lvert c_k \rvert$	$\frac{1}{2}$	$\frac{1}{2\sqrt{2}}$	0	$\frac{1}{2\sqrt{2}}$
$\angle c_k$	0	$-\frac{\pi}{4}$	$Undefined$	$\frac{\pi}{4}$

DISCRETE TIME FOURIER TRANSFORM (SPECTRUM OF A DISCRETE SIGNAL)

Just as in the case of continuous time aperiodic energy signals, the frequency analysis of discrete time aperiodic finite-energy signals involves a Fourier transform of the time domain signal. The discrete time Fourier transform is thus given by:

$$X_D(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}$$

- Recall that $\Theta_{\text{digital}} = \omega_{analog}\, T$

Physically $X(\theta)$ represents the frequency content of the signal $x[n]$. Note that unlike continuous time signals, the discrete time Fourier transform has a frequency range of $-\pi$ to π and is 2π periodic[1]:

$$\begin{aligned} X(\theta + 2\pi k) &= \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta\, +2\pi kn} \\ &= \sum_{n=\infty}^{\infty} x[n]e^{-jn\theta} \\ &= X(\theta) \end{aligned}$$

Since θ is a continuous frequency variable, we can express it as a Fourier series, with its Fourier coefficients being the values of the sequence $x[n]$. We being with:

$$X(\theta)e^{j\theta m} = \sum_{n=\infty}^{\infty} x[n]e^{j\theta\,(m-n)}$$

We then integrate in the interval $[-\pi, \pi]$ and let $N \to \pi$ so that we can interchange the order of summation and integration:

$$\int_{-\pi}^{\pi} X(\theta)e^{jm\theta}\, d\theta = \sum_{n=\infty}^{\infty} x[n] \int_{-\pi}^{\pi} e^{j\theta\,(m-n)}\, d\theta$$

$$= \begin{cases} 2\pi x(m), & m = n \\ 0, & m \neq n \end{cases}$$

$$\therefore\ x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\, d\theta$$

$$x[n] = 1/2\pi \int_{-\pi}^{\pi} x_D(\theta)e^{jn\theta}\, d\theta \quad \overset{DTFT}{\longleftrightarrow} \quad X_D(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}$$

[1] It is possible to show that $X_D(\theta)$ is the sum of the analogue spectrum $X(\theta)|_{\theta=\omega T}$ repeated at every 2π interval.

PROPERTIES OF THE DTFT

LINEARITY

$$ax[n] + by[n] \overset{DTFT}{\longleftrightarrow} aX_D(\theta) + bY_D(\theta)$$

FREQUENCY SHIFT

$$e^{jkn}\, x[n] \overset{DTFT}{\longleftrightarrow} X_D(\theta - k)$$

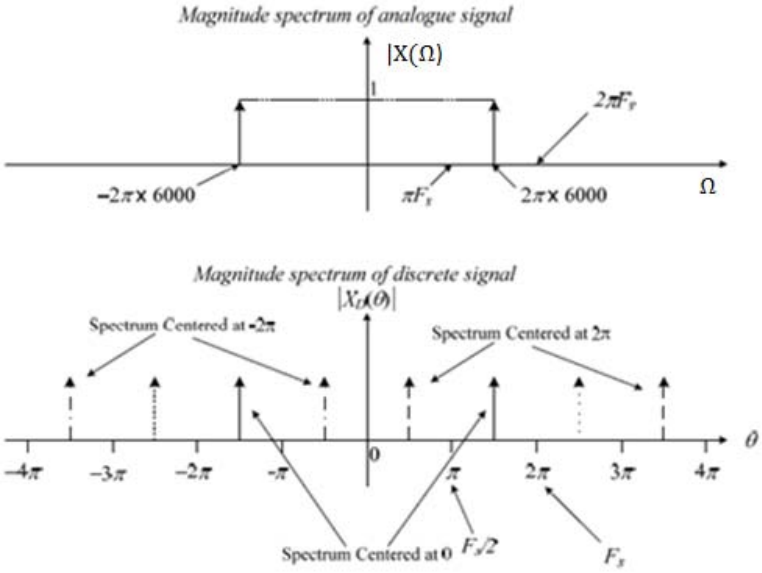
TIME SHIFT

$$x[n - n_0] \overset{DTFT}{\longleftrightarrow} e^{-j\omega n_0} X_D(\theta)$$

Example

If a cosine wave of frequency $f = 6000\ Hz$ is sampled at a rate of $f_s = 8000\ Hz$, sketch the spectrum of the analogue and discrete signals.

Obviously, aliasing will occur in this situation.



DISCRETE FOURIER TRANSFORM (DFT)

It is convenient to map a finite length sequence from the time domain into a finite length sequence of the same length in a different domain ($DFT: \mathbb{C}^N \rightarrow \mathbb{C}C^N$). Let us begin with the DTFT of an aperiodic finite-energy signal $x[n]$ which can also be thought of as a vector:

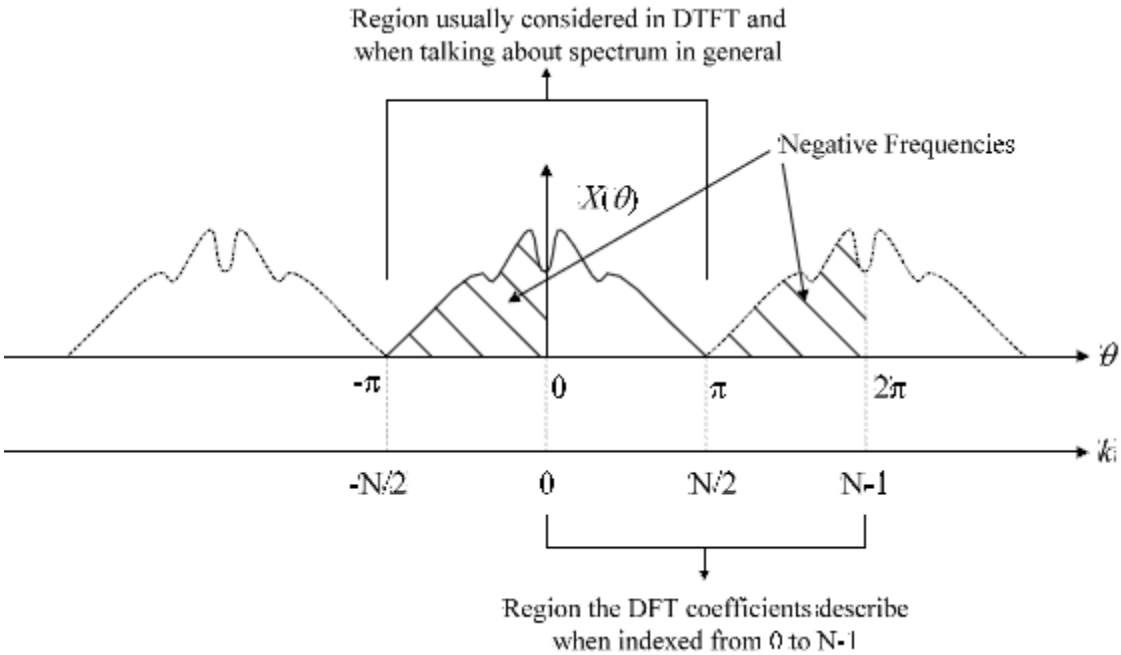
$$X_D(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}$$

Suppose that we sample $X_D(\theta)$ periodically in frequency at a spacing of $\delta\omega$ radians between successive samples. Note that $X_D(\theta)$ is 2π periodic, so only samples in the fundamental frequency range are necessary. If we take N evenly sized samples, then: $\delta\omega = \frac{2\pi}{N}$. Such sampling results in $x[k] = X_D\left(\frac{2\pi k}{N}\right)$:

$$x[k] = \sum_{n=0}^{N-1} x[n]e^{-jn\left(\frac{2\pi k}{N}\right)}$$

Since the frequency samples are obtained by evaluating the Fourier Transform $X(\theta)$ at a set of N discrete frequencies, the relation is defined as the **discrete Fourier transform (DFT)**. The inverse DFT:

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{jn\left(\frac{2\pi k}{N}\right)}$$



Complex Exponential Form of DFT

Often, the complex exponential scalar is used to represent:

$$e^{j\left(\frac{2\pi}{N}\right)} = \omega$$

This indicates that ω_N is one of the N-th root of unity such that $\omega_N^N = e^{j\left(\frac{2\pi N}{N}\right)} = 1$. We thus let $\boldsymbol{\omega}^k$ be the vector with elements consisting of the N-th roots of unity raised to the power of k:

$$\boldsymbol{\omega}^k = \begin{pmatrix} 1 \\ \omega^k \\ \dots \\ \omega^{(N-1)k} \end{pmatrix}$$

If we let $\boldsymbol{\omega}^n[m] = \omega^{kn} = e^{\frac{j2\pi}{N}nm}$ be the m-th vector element, then we can express the DFT in vector form:

$$X[k] = \sum_{n=0}^{N-1} x[n]\omega^{-n}[k]$$

$$\boldsymbol{X} = \begin{bmatrix} X[0] \\ X[1] \\ \dots \\ X[N-1] \end{bmatrix} = \sum_{n=0}^{N-1} x[n]\boldsymbol{\omega}^{-n}$$

DFT as a Linear Transformation

This idea that the DFT can be thought of as a linear transformation comes from the previous relation:

$$\boldsymbol{X} = \sum_{n=0}^{N-1} x[n]\boldsymbol{\omega}^{-n}$$

Note that $X[0] = \sum_{n=0}^{N-1} x[n]$, $X[1] = \sum_{n=0}^{N-1} x[n]e^{\frac{j2\pi n}{N}} = \sum_{n=0}^{N-1} x[n]\omega^{-n}$... We see then that:

$$\underbrace{\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ X[N-1] \end{bmatrix}}_{\boldsymbol{X}} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega^{-1\cdot 1} & \omega^{-1\cdot 2} & \dots & \omega^{-1\cdot(N-1)} \\ 1 & \omega^{-2\cdot 1} & \omega^{-2\cdot 2} & \dots & \omega^{-2\cdot(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{-(N-1)\cdot 1} & \omega^{-(N-1)\cdot 2} & \dots & \omega^{-(N-1)^2} \end{bmatrix}}_{\boldsymbol{\mathcal{F}}} \underbrace{\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix}}_{\boldsymbol{x}}$$

ORTHOGONALITY

The dot product of two vectors $\boldsymbol{\omega}^k$ and $\boldsymbol{\omega}^l$ where k and l are integers is given by:

$$\boldsymbol{\omega}^k \cdot \boldsymbol{\omega}^l = \sum_{n=0}^{N-1} \boldsymbol{\omega}^k[n]\overline{\boldsymbol{\omega}^l[n]} = \sum_{n=0}^{N-1} \omega^{(k-l)n}$$

Note that $\boldsymbol{a} \cdot \boldsymbol{b} = \sum a_i \bar{b}_i$. The last summation is a geometric series of the value $\left(\frac{1-\left(\omega^{(k-l)}\right)^N}{1-\omega^{k-l}}\right)$ and has the values of:

$$\sum_{n=0}^{N-1} \omega^{(k-l)n} = \begin{cases} 0, & k \neq l \\ N, & k = l \end{cases}$$

COMPLEX EXPONENTIAL OF IDFT

Let us take the vector **X** (DFT coefficients) and dot product it with $\boldsymbol{\omega}^{-n}$.

$$X \cdot \omega^{-n} = \sum_{k=0}^{N-1} x[k]\overline{\omega^{-n}[k]} = \sum_{k=0}^{N-1} x[k] e^{\left(\frac{j2\pi}{N}\right)kn}$$

But recall that we can express $x[k]$ as:

$$x[k] = \sum_{m=0}^{N-1} x[m]\, e^{\left(\frac{-j2\pi}{N}\right)km}$$

So:

$$\sum_{k=0}^{N-1} x[k] e^{\left(\frac{j2\pi}{N}\right)kn} = \sum_{k=0}^{N-1} \left[\sum_{m=0}^{N-1} x[m]\, e^{\left(\frac{j2\pi}{N}\right)k(m-n)}\right]$$

$$= \sum_{m=0}^{N-1} x[m] \left[\sum_{k=0}^{N-1} \omega^{k(m-n)}\right]$$

But note that this sum is much like the definition of the dot product with m=n so:

$$= \sum_{m=0}^{N-1} x[m]\,(\boldsymbol{\omega}^m \cdot \boldsymbol{\omega}^n)$$
$$= Nx[n]$$

We thus arrive at the IDFT:

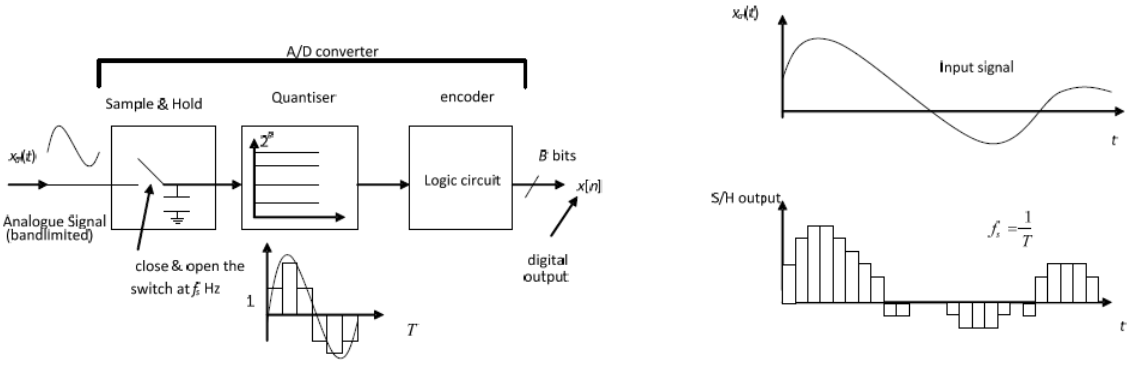
$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[k]\omega^{kn}$$

$$\boldsymbol{X} = \frac{1}{N}\sum_{k=0}^{N-1} x[k]\boldsymbol{\omega}^k$$

ANALOGUE TO DIGITAL CONVERSION

Before any DSP algorithm can be performed, a signal must be converted from its analogue form to a digital form by sampling.

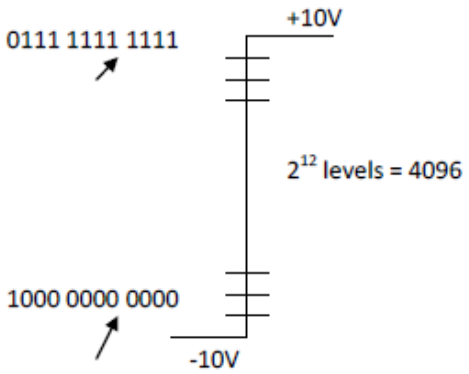
1. The band limited signal must first be sampled and converted into a discrete time signal
2. The amplitude of each sample is quantized into one of 2^B levels, where B is the number of bits used to represent a sample
3. The discrete amplitude levels are represented or encoded into distinct binary words each of length B bits.



Since the signal must be sampled and converted in a discrete time signal, it is usual to use **sample and hold** (S/H) to take a snapshot of the analogue signal every T seconds and to hold that value constant until the next snapshot is obtained.

QUANTISATION, ENCODING & SQNR

Each analogue sample is assigned one of 2^B values when quantization occurs. This quantization process however introduces irreparable errors. This is because an analogue signal could have an amplitude between two quantized level. When this occurs, the D/A converter approximates its amplitude as being the level closest to the actual amplitude.



The **resolution of the quantization** (quantization step size) is given by the full scale range (V) divided by the number of gaps between two successive levels given by $2^B - 1$:

$$\Delta V = \frac{V}{2^B - 1} \approx \frac{V}{2^B}$$

We thus find that the maximum quantization error occurs when an analogue signal has an amplitude in the middle of two successive levels. Hence:

$$e_{max} = \pm \frac{\Delta V}{2}$$

The quantization error is assumed to be random and uniformly distributed so:

$$\int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} P \, de = 1 \rightarrow P = \frac{1}{\Delta V}$$

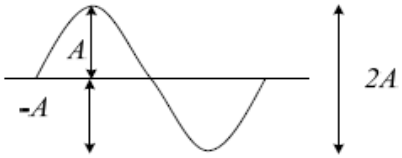
We can thus find the quantization noise power or variance:

$$\sigma_e^2 = \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 P(e) de = \frac{1}{\Delta V} \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 de$$

$$= \frac{\Delta V^2}{12}$$

This allows us to find the signal-to-quantisation noise power ratio. For a sine wave, the average signal power is $\frac{A^2}{2}$ and the quantization step size is approximately given by $\Delta V = \frac{2A}{2^B}$. Hence:

$$\Delta V \approx \frac{2A}{2^B}$$



$$SQNR = 10 \log\left(\frac{\left(\frac{A^2}{2}\right)}{\frac{\Delta V^2}{12}}\right)$$

$$= 10Log\left(\frac{\left(\frac{A^2}{2}\right)}{\frac{\left(\frac{2A}{2^B}\right)^2}{12}}\right)$$

$$= 6.02B + 1.76\ dB$$

$$\Delta V = \frac{V}{2^B - 1} \approx \frac{V}{2^B}$$

$$e_{\max} = \pm\frac{\Delta V}{2}$$

$$\sigma_e^2 = \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 P(e) de = \frac{1}{\Delta V} \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 de = \frac{\Delta V^2}{12}$$

$$SQNR = 6.02B + 1.76\ dB$$

NYQUIST SAMPLING THEOREM

If the highest frequency contained in an analogue signal $x(t)$ is F_{max} and the signal is sampled at a rate $F_s \geq 2F_{max}$ then $x(t)$ can be exactly recovered from an infinite sequence of samples using an interpolation function.

$$F_s \geq 2F_{max}$$

Quick Proof:

Recall that for a discrete signal:

$$-\pi \leq \omega \leq \pi, \quad -\frac{1}{2} \leq f \leq \frac{1}{2}$$

Alternatively we find after substitution of the variables into the last equation:

$$F_{max} \leq \left|\frac{F_s}{2}\right|, \quad \Omega \leq |\pi F_s|$$

Since the highest frequency in a discrete time signal is $\omega = \pi, f = \frac{1}{2}$ then the highest value of F and Ω would be $\frac{F_s}{2}$, πF_s respectively. This implies that sampling introduces ambiguity (the aliasing effect), since the highest frequency in a continuous time signal that can be uniquely distinguished is $\frac{F_s}{2}$.

The next section shows the proof of the ability to recover the original signal.

Proof of Recoverability:

The following proof requires the knowledge of the following properties of the Dirac delta (impulse) functions $\delta(t)$

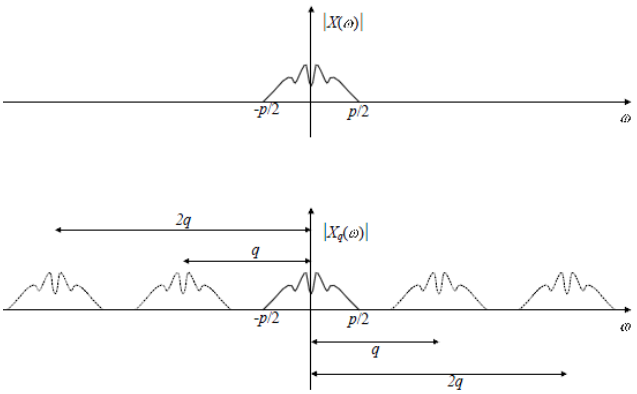
$$f(x) \cdot \delta(x-a) = f(a) \cdot \delta(x-a)$$

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = f(a)$$

$$f(x) * \delta(x-a) = f(x-a)$$

1. Consider a bandlimited signal $x(t)$.

$$\mathcal{F}[x(t)] = X(\omega) = 0, \quad |\omega| > \frac{p}{2}$$



2. We periodise the signal with a period q since sampling a continuous time signal with sampling rate F_s produces a discrete time signal whose frequency spectrum is the periodic replication of the original signal, with a replication period of F_s. The reconstruction (without proper filtering) of such a signal will also result in images [2]

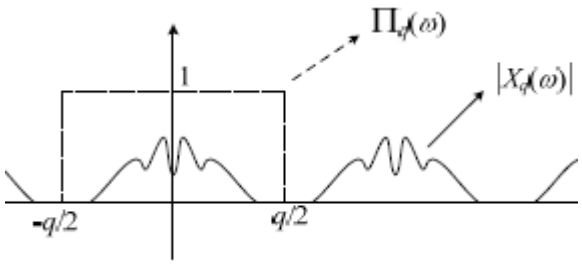
$$\begin{aligned} X_q(\omega) &= \sum_{k=-\infty}^{\infty} X(\omega - kq) \\ &= \sum_{k=-\infty}^{\infty} X(\omega) * \delta(\omega - kq) \\ &= X(\omega) * \sum_{k=-\infty}^{\infty} \delta(\omega - kq) \end{aligned}$$

Let $\rho_q(\omega) = \sum_{k=-\infty}^{\infty} \delta(\omega - kq)$

$$X_q(\omega) = X(\omega) * \rho_q(\omega),$$

For $q > p$, the repeated images do not overlap and the original spectrum $X(\omega)$ can be recovered from the periodised version, by multiplying $X_q(\omega)$ with the rectangular function:

$$\Pi_q(\omega) = \begin{cases} 1, & |\omega| \le \frac{q}{2} \\ 0, & |\omega| \ge \frac{q}{2} \end{cases}$$



We thus get:

$$\begin{aligned} X(\omega) &= \Pi_q(\omega) \cdot X_q(\omega) \\ &= \Pi_q(\omega) \cdot \big(X(\omega) * \rho_1(\omega)\big) \quad q > p \end{aligned}$$

We take the inverse Fourier Transform:

$$\begin{aligned} x(t) &= \mathcal{F}^{-1}[X(\omega)] \\ &= \mathcal{F}^{-1}\big[\Pi_q(\omega)\big] * \mathcal{F}^{-1}[X(\omega) * \rho_1(\omega)] \\ &= \mathcal{F}^{-1}\big[\Pi_q(\omega)\big] * 2\pi\big(\mathcal{F}^{-1}[X(\omega)] \cdot \mathcal{F}^{-1}\big[\rho_q(\omega)\big]\big) \end{aligned}$$

3. To work out the inverse Fourier transform of each term we note:

$$\mathcal{F}^{-1}[X(\omega)] = x(t) \qquad (1)$$

$$\begin{aligned} \mathcal{F}^{-1}\big[\Pi_q(\omega)\big] &= \frac{1}{2\pi} \int_{-\frac{q}{2}}^{\frac{q}{2}} e^{j\omega t}\, d\omega \\ &= \frac{1}{2\pi jt}\left[e^{\frac{jqt}{2}} - e^{\frac{-jqt}{2}}\right] \end{aligned}$$

[2] When sampling at f_s, we cannot distinguish between the sampled values of a signal of f_0 Hz and a signal of $f_0 + kf_s$ Hz.

$$= \frac{1}{\pi t}\sin\left(\frac{qt}{2}\right)$$
$$= \frac{q}{2\pi}\frac{sin\left(\frac{qt}{2}\right)}{\frac{qt}{2}}$$
$$= \frac{q}{2\pi}sinc\left(\frac{qt}{2}\right) \qquad (2)$$

$\rho_q(\omega)$ is a periodic function ($\omega = 2\pi/q$) that can be represented by a Fourier series:

$$\rho_q(\omega) = \sum_{k=-\infty}^{\infty} c_k e^{\frac{jk2\pi}{q}\omega}$$

Note that ω is now the independent variable. So the Fourier coefficients are given by:

$$c_k = \frac{1}{q}\int_{-\frac{q}{2}}^{\frac{q}{2}} \rho_q(\omega) e^{-jk\left(\frac{2\pi}{q}\right)\omega}\, d\omega$$

The Dirac Comb property: $\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = f(a)$ can be used, noting that a=0 leaving behind:

$$c_k = \frac{1}{q}$$

Thus:

$$\rho_q(\omega) = \frac{1}{q}\sum_{k=-\infty}^{\infty} e^{\frac{jk2\pi}{q}\omega}$$

$$\mathcal{F}^{-1}[\rho_q(\omega)] = \frac{1}{q}\sum_{k=-\infty}^{\infty} \mathcal{F}^{-1}\left[e^{\frac{jk2\pi}{q}\omega}\right]$$

Since $\mathcal{F}^{-1}[e^{-j\omega t_0}] = \delta(t-t_0)$

$$\mathcal{F}^{-1}[\rho_q(\omega)] = \frac{1}{q}\sum_{k=-\infty}^{\infty} \delta\left(t + k\left(\frac{2\pi}{q}\right)\right)$$

Let $T = \frac{2\pi}{q}$, $k = -n$

$$\mathcal{F}^{-1}[\rho_q(\omega)] = \frac{T}{2\pi}\sum_{k=-\infty}^{\infty} \delta(t-nT) \qquad (3)$$

4. Putting all of part 3 components together into the equation for part 2

$$x(t) = \frac{1}{T}sinc\left(\frac{\pi t}{T}\right) * 2\pi\left(x(t)\cdot\frac{T}{2\pi}\sum_{-\infty}^{\infty}\delta(t-nT)\right)$$

$$= sinc\left(\frac{\pi t}{T}\right) * \left(\sum_{n=-\infty}^{\infty} x(nT)\delta(t-nT)\right)$$

Note that $x(t)\delta(t-a) = x(a)\delta(t-a)$ then we have $x(t)\delta(t-nT) = x(nT)\delta(t-nT)$. This is rather significant since $x(nT)$ are the values of the discrete signal $x[n]$, obtained by taking evenly spaced samples every T seconds. If we replace $x(nT)$ with $x[n]$ then $x[n]$ is no longer a function of T so:

$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\left(sinc\left(\frac{\pi t}{T}\right) * \delta(t-nT)\right)$$

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \cdot sinc\left(\frac{\pi(t-nT)}{T}\right)$$ [3]

In this case, the sinc function is independent of $x(t)$. This indicates that we can recover the original signal $x(t)$ from a discrete signal $x[n]$ as long as the original expression holds:

$$X(\omega) = \Pi_q(\omega) \cdot \Big(X(\omega) * \rho_q(\omega)\Big), \quad q > p$$

Since the original signal is bandlimited, $\frac{p}{2}$ corresponds to the maximum frequency (in rad/s) present in the signal so:

$$\frac{p}{2} = 2\pi F_{max}$$

$$p = 4\pi F_{max}$$

We previously let $T = \frac{2\pi}{q}$ hence:

$$q = \frac{2\pi}{T} = 2\pi F_s$$

Then:

$$2\pi F_s > 4\pi F_{max}$$

$$F_s > 2F_{max}$$

Which is the Nyquist Sampling Theorem. Now we define the Nyquist rate to be F_s .

Example

What is the Nyquist rate for this analog signal?

$$x(t) = 3cos50\pi t + 10sin300\pi t - cos100\pi t$$

$$F_{max} = 150\ Hz$$

Then:

$$F_s = 300\ Hz$$

[3] Whittaker-Shannon Interpolation formula

Example

Given a signal $g(t) = sinc(200\pi t)$, find the Nyquist rate and the Nyquist interval for the signal.

Now we need to find $\mathcal{F}\{sinc(200\pi t)\}$ which is by no means an easy task. We thus first start off by guessing a solution and working backwards:

$$\mathcal{F}[\Pi(t)] = \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} Ae^{-j\omega t}\ dt = A\tau sinc\left(\frac{\omega\tau}{2}\right)$$

By the duality principle (replace t by –t and interchanging t and ω) then:

$$\mathcal{F}\left[A\tau sinc\left(\frac{\tau t}{2}\right)\right] = 2\pi\Pi(\tau)$$

We thus find that:

$$\mathcal{F}[sinc(200\pi t)] = 2\pi\Pi(\tau)$$

Where $\frac{\tau}{2} = 200\pi$

$$\tau = 400\pi = T = 2\pi f_{max}$$

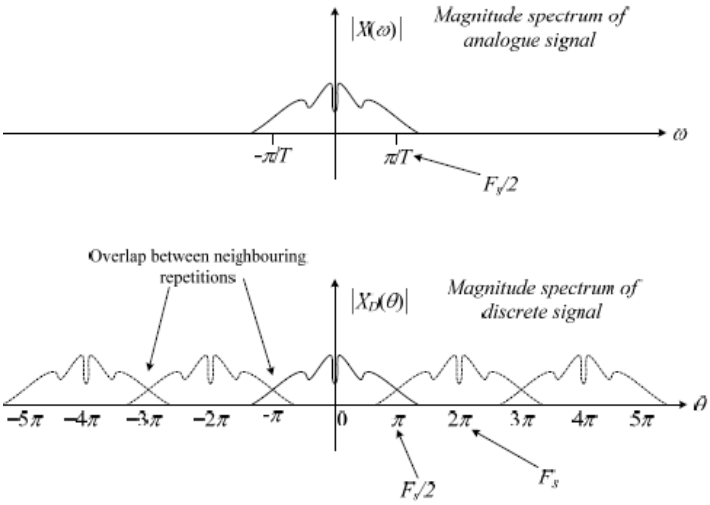
$$\therefore f_{max} = 200\ Hz$$

$$Nyquist\ rate: \frac{f_s}{2} = f_{max} = 200Hz$$

$$Nyquist\ Interval: \frac{1}{f_s} = \frac{1}{200} = 0.05s$$

ALIASING

If the sampling condition is not satisfied: $f_s \geq 2f_{max}$, then neighbouring repetitions of the spectra overlap (and add) to distort the signal. This overlap means that any frequency component above $\frac{f_s}{2}$ is indistinguishable from a lower frequency component, called an alias, associated with one of the copies. Thus when $X_q(\omega)$ and multiplied by the rectangular function $\Pi_q(\omega)$, the result does not return the spectrum of the original signal.



The overlap in the spectrum causes part of it to fold over and alias back into itself. This is known as aliasing.

Example

Consider $x(t) = 3cos2000\pi t + 5sin6000\pi t + 10cos12000\pi t$.

a) What is the Nyquist rate for this signal?
b) Assume that $F_s = 5000\ Hz$. What is the discrete time signal obtained after sampling?
c) What is the analog signal $y(t)$ we can reconstruct from the samples if we use ideal interpolation?

a) The frequencies that exist in the analog signal are 1 kHz, 3 kHz, and 6 kHz using $\cos(2\pi Ft) = \cos(\Omega t)$. The Nyquist rate should thus be $F_s = 12\ kHz$

b) $$x[n] = 3\cos 2\pi\left(\frac{1}{5}n\right) + 5\sin 2\pi\left(\frac{3}{5}n\right) + 10\cos 2\pi\left(\frac{6}{5}n\right)$$
$$= 3\cos\left(\frac{2\pi}{5}n\right) + 5\sin\left(-\frac{4\pi}{5}n\right) + 10\cos\left(\frac{2\pi}{5}n\right)$$
$$= 13\cos\left(\frac{2\pi}{5}n\right) + 5\sin\left(-\frac{4\pi}{5}n\right)$$

This aliasing effect can be verified since the frequencies will fold:

$$F_2^{'} = 3 - 5 = -2\ kHz,\ \ F_3^{'} = 6 - 5 = 1\ kHz$$

c) We let $n = \frac{t}{T} = 5000t$ and note that only the 1 kHz and 2 kHz components remain.
$y(t) = 13\cos(2000\pi t) - 5\sin(4000\pi t)$

FFT ALGORITHM & ZERO PADDING

The Fast Fourier Transform (FFT) is simply a mathematical technique to accelerate the calculation of the DFT and IDFT. It operates on signal vectors of length that is a power of 2 usually. In such cases, it is common to add zeros at the end of the signal to make up the required length. This technique is called **zero padding**.

One can show that the zero valued elements contribute nothing, but only act to decrease the frequency spacing from $\frac{2\pi}{N}$ to $\frac{2\pi}{M}$ for M>N, which are the lengths of the zero padded and original signal respectively.

The zero padding technique gives us a high density spectrum and provides for a better displayed version for plotting. It however does not give us a high resolution spectrum because no new information is added to the signal, only additional zeroes.